



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2028	B.TECH	IIT Kharagpur	7.45 / 10
2024	Board of Intermediate Education	Sri Chaitanya College	98.6%
2022	Board of Secondary Education, Andhra Pradesh	VSSC Holy Cross School	93.8%

PROJECTS

CreditGaurd - Real Time Fraud Detection and Anomaly Intelligence Engine

- Built a hybrid fraud detection system pairing supervised classification with a **PyTorch autoencoder** for anomaly Detection
- Applied **SMOTE** and cost-sensitive learning to tackle class imbalance, boosting fraud recall while cutting false positives
- Engineered a low latency **C++(STL based)** scoring module for **sub-millisecond** edge inference, bridged to python via **pybind11**
- Tracked experiments with **Mlflow** stored data in **MySQL**, and served predictions via **Fast API** and **Docker** containers
- Built a **Power BI** dashboard monitoring fraud trends and model drift for real-time analytics and stakeholder reporting

PredictAI - Predictive Maintenance for industrial Equipment using deep learning

- Built an **LSTM**-based deep learning model to predict equipment failure from noisy, **multivariate sensor** time-series data
- Engineered time-series features (rolling statistics, **FFT**-based frequency components) to capture early degradation signals
- Built the feature pipeline with NumPy, Pandas, and SciPy to process raw multivariate sensor signals at scale
- Used **Mlflow** for experiment tracking and **DVC** for dataset versioning across iterative training run
- Deployed **Streamlit** dashboard simulating real-time sensor streams with failure alerts and **remaining-useful-life** estimates

PulseBoard - Multilingual Customer Feedback Intelligence and BI Platform

- Built an **NLP pipeline** using **Hugging Face transformers** and **NLTK** to classify **sentiment**, detect **emotion** and theme extraction
- Designed a **MySQL**-backed **data warehouse** to store feedback and aggregate sentiment metrics across product lines and regions
- Exposed real-time inference via **FastAPI** and automated Excel report generation for non technical stakeholders
- Built Interactive **Power BI** and Tableau dashboards surfacing sentiment trends and **churn-risk** signals for product and cx teams

AgentIQ - Agentic RAG Assistant with Self-correcting Retrieval and Tool Use

- Built a multi-step agent with **LangGraph** orchestrating retrieval, reasoning, and tool calls over a **hybrid vector+keyword index**
- Implemented **Corrective RAG** - Agent grades chunk relevance and re-queries or falls back to web search when confidence is low
- Exposed external tools (calculator, SQL, Web search) to the agent via **MCP**, letting it choose tools dynamically per query
- Engineered a **RAGAS** - Based Evolution harness measuring faithfulness, answer relevance and context precision across test set

SKILLS AND EXPERTISE

- **Programming Languages** : Python | C | C++ | MySQL
- **AI/LLM Technologies** : LangChain | LangGraph | Hugging Face | Retrieval-Augmented Generation(RAG) | Prompt Engineering | MCP
- **Libraries** : NumPy | Pandas | Scikit-Learn | NLTK | TensorFlow | PyTorch
- **Framework and Tools** : Streamlit | FastAPI | Docker | Github Actions(CI/CD) | Git | Mlflow | Rest APIs | Jupyterlab | VS code
- **Data Visualization** : Power BI | Tableau | Matplotlib | Seaborn | Microsoft Excel

COURSEWORK INFORMATION

- **Mathematics**: Advanced Calculus | Partial Differential Equations | Integral Transforms and Applications | Probability and Statistics
- **Problem Solving**: Programming and Data Structures | Essentials of Machine Learning | Operations Research

CERTIFICATIONS

- Completed a structured **Prompt Engineering** course covering Advanced AI prompting strategies and practical use cases
- Certified in **AutoCAD** and **Ansys** through 1Stop.ai , covering CAD modelling, 3D Design and engineering simulation

POSITIONS OF RESPONSIBILITY

Sub-Head, Web Team | Shaurya, IIT Kharagpur

- Built the official Shaurya website with **React.js** and **Tailwind CSS** in a 6-member web team
- Developed the **Team Page** and multiple event facing pages with **responsive design**
- Managed end-to-end **transportation logistics** for participants and guests during the fest
- Served as **Point of Contact** across teams, Streamlining inter-team communication in real time

EXTRA CURRICULAR ACTIVITIES

- Won **silver medal** in **Cricket** at IIT Kharagpur's **Inter-Hall** sports competition(General Championship)
- Led Team to Victory at **Kshitij BGMI tournament**, captaining a squad through bracket play to secure the **championship title**
- Honed Team Coordination and court skills through **NSO volleyball** coaching at IIT Kharagpur